HARMONY TUTOR

The intelligent tonal-harmony checker and multi-voice writing

USER MANUAL

Version 1. 2 — June 2026
macOS · Windows

Contents

1. Introduction

1.1 What is Harmony Tutor

Harmony Tutor is a SATB (Soprano, Alto, Tenor, Bass) music editor with automatic, real-time harmonic analysis. It is designed for harmony students and teachers, musicians and composers who want to check and improve their four-part writing exercises.

The application analyses each chord the moment it is entered, identifies voice-leading errors, recognises cadences and modulations, and provides a didactic explanation for every flag. It is currently the only SATB editor with real-time harmonic analysis available.

1.2 Who it is for

- **Conservatory and music-school students:** immediate checking of exercises, understanding of mistakes, self-study.
- **Harmony teachers:** a fast correction tool, teaching support, time saved on recurring errors.
- **Self-taught musicians:** learning guided by practice and feedback.
- **Composers and arrangers:** voice-leading checking and harmonic analysis in a classical context.

1.3 Core principles

- **Spelling-First:** analysis is based on note names (spelling), not only on MIDI numbers. This guarantees correct distinction of enharmonic intervals and reliable recognition of augmented-sixth chords (It6, Fr6, Ger6).
- **Educational correction:** every violation shows a rule code, an expandable description and a practical tip for fixing it.
- **International academic standard:** nomenclature follows the academic standard, supporting both Roman numerals (functional figuring) and modern chord symbols (Cmaj7, Dm, G7/F, etc.).

1.4 System requirements

Operating system	Minimum requirements
macOS	macOS 10.15 (Catalina) or later
Windows	Windows 10 (64-bit) or later

The application does not require an internet connection. Recommended resolution: 1280×800 pixels or higher.

2. Installation and first launch

2.1 macOS

1. Download the Harmony Tutor-mac.dmg file from the website.
2. Open the DMG and drag “Harmony Tutor” into the Applications folder.

2.2 Windows

1. Download Harmony Tutor.win.exe from the website and run the installer.
2. If the “Windows protected your PC” (SmartScreen) warning appears, click “More info” and then “Run anyway”.
3. Follow the wizard. The app will be available in the Start menu.

2.3 First launch

On opening, Harmony Tutor presents an empty project in C major, 4/4 time, with 4 measures. The interface is in dark mode.

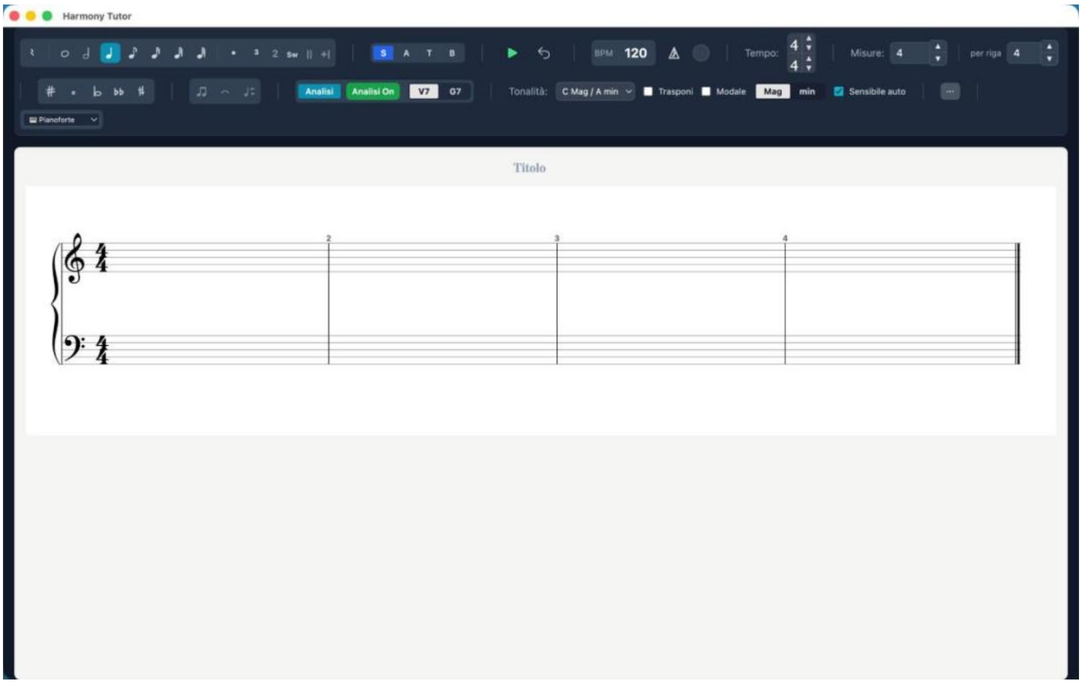


Figure 1 — First launch: empty project in C major, 4/4, 4 measures

2.4 New project (initial tracks and templates)

Creating a new project (File ‣ New, or the New action in the side menu) opens the New project dialog, where you set up the starting tracks before the project is created. You can always add or remove tracks later from the Mixer.

- SATB staff — the “SATB (4-voice choir)” checkbox shows or hides the four-part vocal staff. Uncheck it for an accompaniment- or percussion-only project.
- Accompaniment tracks — pick an instrument from the drop-down and click “+ Instrument” to add a pitched track; “+ 🥁 Drums” adds a percussion track. The chosen tracks are listed below; ✕ removes one.
- Templates — “Save current configuration…” stores the current set of tracks under a name; click a saved template to load it into the selection, 🗑 to delete it. Templates are kept locally and are offered for every new project.
- Create project / Cancel — confirm the configuration to create the project, or close the dialog without creating it.

3. The interface

3.1 Overview

- **Toolbar (top):** controls for voice, durations, accidentals, key, meter, playback and instrument.
- **Central editor:** the staff with the SATB voices, harmonic labels (Roman numerals) below the staff and chord symbols above (toggleable).
- **Analysis panel (right):** list of violations, severity filters, rule search, explanation of harmonic labels.

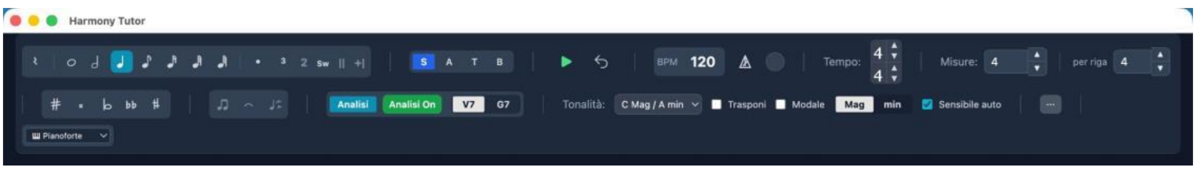


Figure 2 — The main toolbar

3.2 Display modes

The editor supports several layouts, selectable from the Preferences panel (Editor tab) or the side menu. Switching layout is instantaneous: the notes are unchanged.

Mode	Description	Staves
Grand Staff (default)	Treble clef + bass clef	2
Treble clef	A single staff	1
Old SATB clefs	Four staves with old clefs (S, A, T, B)	4
Close score	Grand Staff with reduced spacing	2
Open score	Separate staves with wide spacing	4

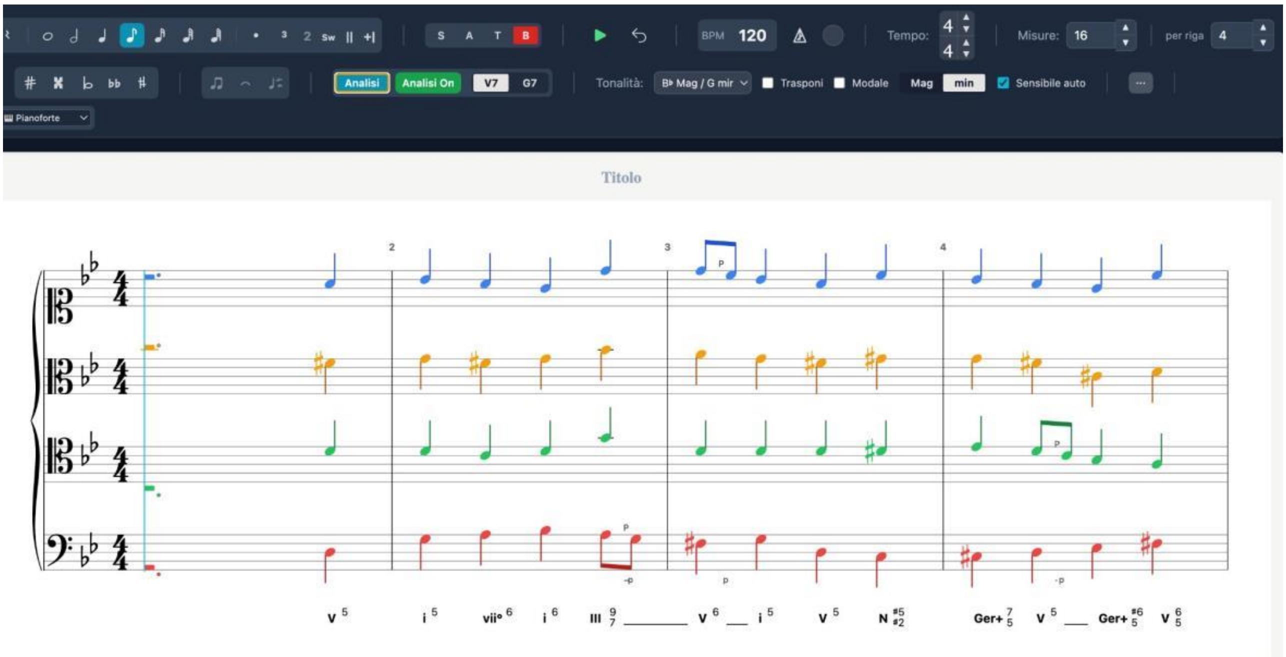


Figure 3 — SATB layout with old clefs, coloured voices, augmented sixths and Neapolitan

3.3 Side menu

The "..." button in the toolbar opens the side menu, which gathers the layout, format and MIDI controls.

- **Layout:** Grand Staff (2 staves), Treble clef (1 staff), Old SATB clefs (4 staves), Close score.
- **Format:** Page (portrait) or Landscape. Affects the on-screen layout and the PDF export.
- **MIDI:** enable/disable MIDI output and select the destination device (Internal Audio, IAC Driver Bus, Logic Pro Virtual In or other connected devices).
- **Step Input:** when active, notes are entered one at a time from the connected MIDI keyboard; the position advances automatically to the next beat.

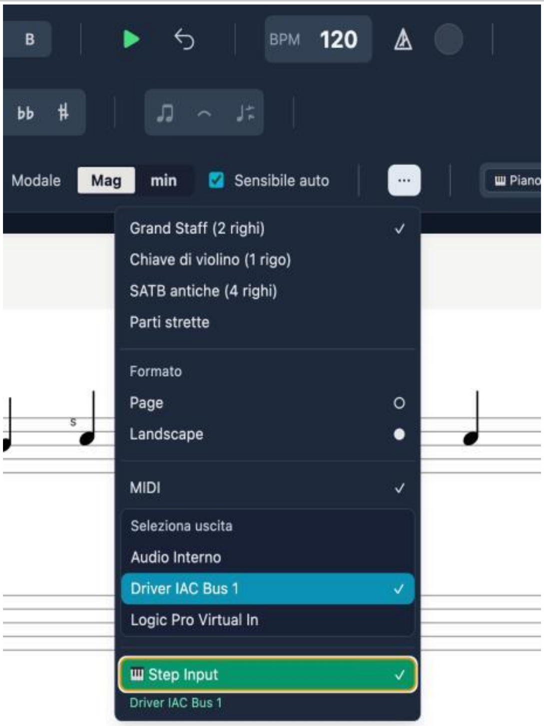


Figure 4 — Side menu: layout, format, MIDI and Step Input

3.4 Voice colours

By enabling “Voice colours (BTAS)” in Preferences → Editor, each voice is shown in a distinct colour: Soprano in blue, Alto in orange, Tenor in green, Bass in red.

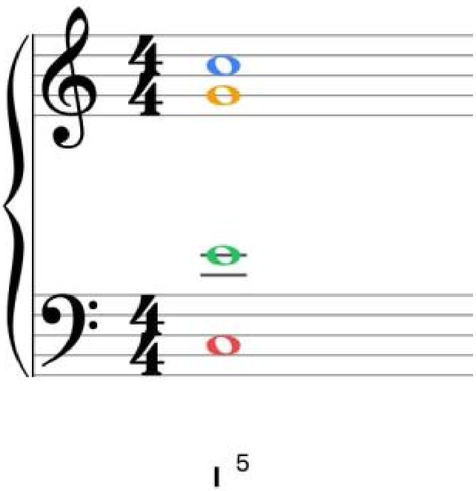


Figure 5 — Coloured voices: S (blue), A (orange), T (green), B (red)

3.5 Voice-leading indicators

On the staff, coloured dashed lines connect the notes between one chord and the next:

- **Green lines:** permitted or tolerated motion, or recognised cadences (PAC, IAC, HC, plagal). They appear only where the engine detects a meaningful pattern.
- **Orange lines:** a warning (situation to check).

- **Red lines:** a serious error (parallel fifths, parallel octaves, voice crossing, etc.).

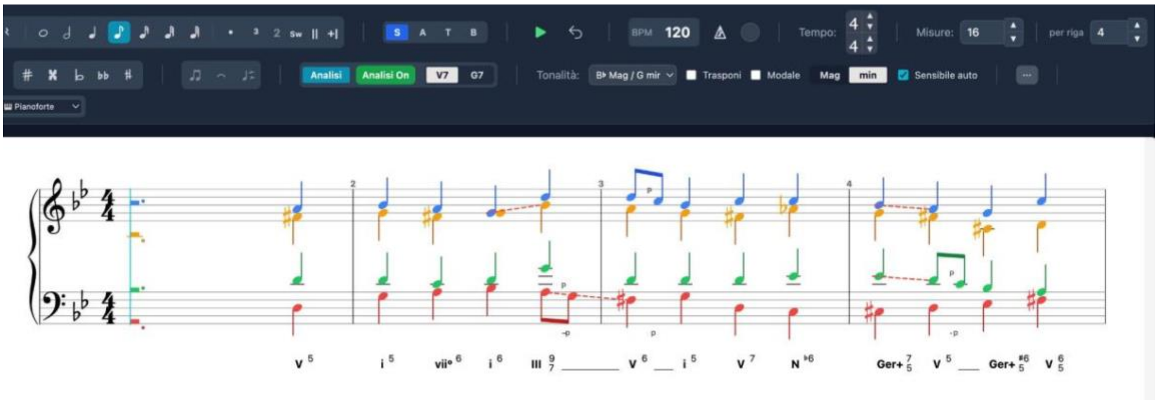


Figure 6 — Voice-leading with green, orange and red lines, Ger+ and Neapolitan

3.6 Chord symbols

In addition to Roman numerals, modern chord symbols (C, Am, D7/F#, Dm7/F, G7, etc.) can be shown above the staff. Toggle them from the toolbar selector or from Preferences → Analysis → “Show chord symbols”.

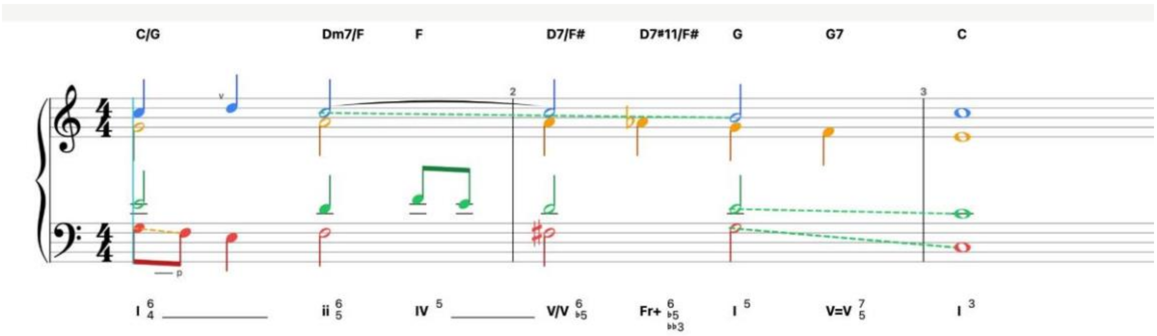


Figure 7 — Dual display: chord symbols above + Roman numerals below, with a French augmented sixth (Fr+)

3.7 Listening to single voices

To hear a single voice in isolation, double-click its voice button (S, A, T or B). Can be enabled on several voices. Double-click again to disable.

Alternatively, more appropriately, by going to the mixer panel accessible from the toolbar, you can solo, mute, or adjust the volume of any track.

More precisely: a double-click on a toolbar S/A/T/B button toggles SOLO for that voice (you can solo several at once). In the Mixer, every voice and ACC-track strip has its own M (mute) and S (solo) buttons, with unified semantics across voices and tracks: as soon as anything is soloed — a voice or a track — only the soloed channels sound, while mute always silences. This is the most flexible way to isolate or compare lines while writing.

3.8 Vocal ranges

Voice	Comfortable range	Extended range
Soprano	C4 – F5	B3 – A5
Alto	F3 – C5	E3 – D5
Tenor	A2 – E4	A2 – A4
Bass	E2 – B3	D2 – C4

3.9 Incomplete-measure warnings (⚠ toggle)

Measures that do not reach the duration required by the meter are flagged with a red rectangle. The warning can now be hidden:

- Click directly on the red overlay on the measure.
- **⚠ button in the toolbar:** red = warnings visible, grey = warnings hidden.

> ***Note:*** *The setting is not persistent: it resets to "visible" each time the file is opened.*

4. Entering notes

4.1 Voice and duration selection

Action	Shortcut	Effect
Cycle voice	V	B → T → A → S → B
Whole note	1	4 beats
Half note	2	2 beats
Quarter note	3	1 beat
Eighth note	4	1/2 beat
Sixteenth note	5	1/4 beat
Thirty-second note	6	1/8 beat
Sixty-fourth note	7	1/16 beat

4.2 Accidentals

Shortcut	Accidental	Effect
#	Sharp (♯)	+1 semitone
b	Flat (♭)	−1 semitone
n	Natural (♮)	Cancel
## (double)	Double sharp	+2 semitones
bb (double)	Double flat	−2 semitones

4.3 Modifiers

Key	Effect
.	Augmentation dot (duration ×1.5)
R	Toggle between note and rest
L	Toggle the tie on the current note
Option/Alt + F	Fermata — see 4.3a

4.3a Fermata

Applies a suspension of metronomic time to a note or rest, lengthening its actual duration in playback (default 2×). The playhead visually freezes and the following events shift accordingly.

1. Select the note or rest.
2. Press Option+F (Mac) or Alt+F (Windows).
3. The fermata appears above the note on the staff.

> ***Note:*** *To remove the fermata: select the note and press Option/Alt+F again.*

4.4 Ghost note and entry

Entering notes with the mouse requires the **Ctrl** (Windows) or **Cmd** (Mac) modifier key, to prevent accidental clicks.

1. **Select the voice** you want (S, A, T or B) from the toolbar or with the V key.
2. **Position yourself on the correct staff** (treble clef for Soprano/Alto, bass clef for Tenor/Bass). If voice and staff are inconsistent, the system blocks entry and shows no note.
3. **Hold Ctrl/Cmd:** the semi-transparent “ghost note” appears as a guide.
4. **Click** on the staff to insert the note definitively.

Without holding Ctrl/Cmd, the mouse acts only as a pointer and selection tool.

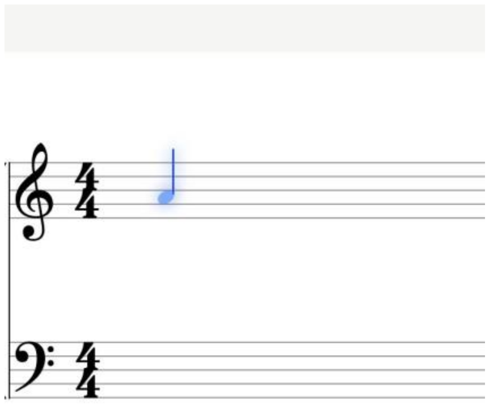


Figure 8 — Ghost note: preview before insertion

4.5 Selecting and editing

Action	Shortcut
Select note	Click
Multi-selection	Shift + Click
Delete	Backspace / Delete
Transpose ±1 semitone	↑ / ↓
Transpose ±1 octave	Alt + ↑ / ↓
Undo / Redo	Cmd/Ctrl + Z / Shift+Z

4.6 Manual ornament marking

With the T panel open, the “Ornament marking” section lets you force the ornament type. Shortcuts use the Option (Mac) or Alt (Windows) key:

Shortcut	Type
Option+H	Structural (chord tone)
Option+P	Passing tone
Option+V	Neighbor tone

Shortcut	Type
Option+A	Appoggiatura
Option+N	Anticipation
Option+S	Escape tone
Option+R	Suspension / Retardation
Option+O	Ornamental (figuration)
Option+C	Changing tone

> ***Note:*** *On Mac the Control key is reserved for toggling note colours in the voices.*

4.7 Voice reassignment

Notes that have already been written can be moved from one voice to another by reusing the toolbar S/A/T/B buttons, without rewriting them. The feature works both in SATB editing and on “Grand staff with voices” ACC tracks.

1. Select the notes (rectangle selection is supported).
2. Click the destination voice (S, A, T or B) in the toolbar: the notes move to that voice.
3. The clef is recomputed from the voice: for example S → Tenor/Bass moves correctly to the lower staff.

Behaviour:

- **Collisions → chord:** if a note already exists at that position in the destination voice, the two become a chord (nothing is lost).
- **Automatic rests:** the source voice fills in rests where the moved notes used to be.
- **Undoable:** the operation is reverted with Cmd/Ctrl+Z.

> ***Note:*** *With no active selection, the S/A/T/B buttons keep their usual function of choosing the entry voice. When a selection is present, the tooltip changes to “move the selected notes here”.*

5. Measure-edit panel (T)

Pressing T with the playhead on a measure opens the side panel with the editing tools.

5.1 Modulation / Tonicisation

Sets a key change from the current measure. Select the new tonic and mode (Maj/min), then “Apply context”.

5.2 Measure

- **Delete measure:** removes the current measure and shifts everything after it back.
- **Repeat:** inserts repeat barlines (open, close, double).
- **Meter change:** applies a meter change from the current measure.

5.3 Harmony override

Manually forces the harmonic label with three fields: Roman (e.g. I, V/vi, Ger+), Figure (e.g. 6/5) and Chord symbol (e.g. D7/F#).

5.4 Move to staff

Moves a note between the treble and bass staff with Option+arrow up/down (Mac) or Alt+arrow (Windows). Allows mixed open/close-score writing on the same system. “Restore default staff” returns the note to the staff assigned by its voice.

Figure 9 — The T panel: modulation, override, repeat, delete measure

6. Key and meter

6.1 Key selection

All 24 major and minor keys. Global change from the toolbar or per single measure from the T panel.

6.2 Minor mode

- **Natural (aeolian):** natural minor scale, with no raised leading tone.
- **Harmonic:** raised 7th degree to create the leading tone (V–i).

The “Auto leading tone” flag in the toolbar automatically inserts the leading-tone accidental in minor mode: the 7th degree is written already altered, without doing it on every note.

6.3 Modulations

- **Automatic recognition:** cadential patterns tested against all candidate tonics.
- **Manual marking:** via the T panel.
- **Look-ahead tonicisations:** the engine looks up to 2 measures ahead.

6.4 Meter and BPM

Default meter: 4/4. Available: 2/4, 2/2, 3/4, 3/8, 4/4, 5/4, 6/8, 7/8, 12/8 and any custom meter. Meter changes from the T panel. Metronome toggled with K.

7. Harmonic analysis

7.1 How it works

Harmony Tutor analyses the notes in real time through an 8-block pipeline. The result is a harmonic label (Roman numeral) below the staff for each chord and/or modern chord symbols above the staff, toggleable from the toolbar selector.

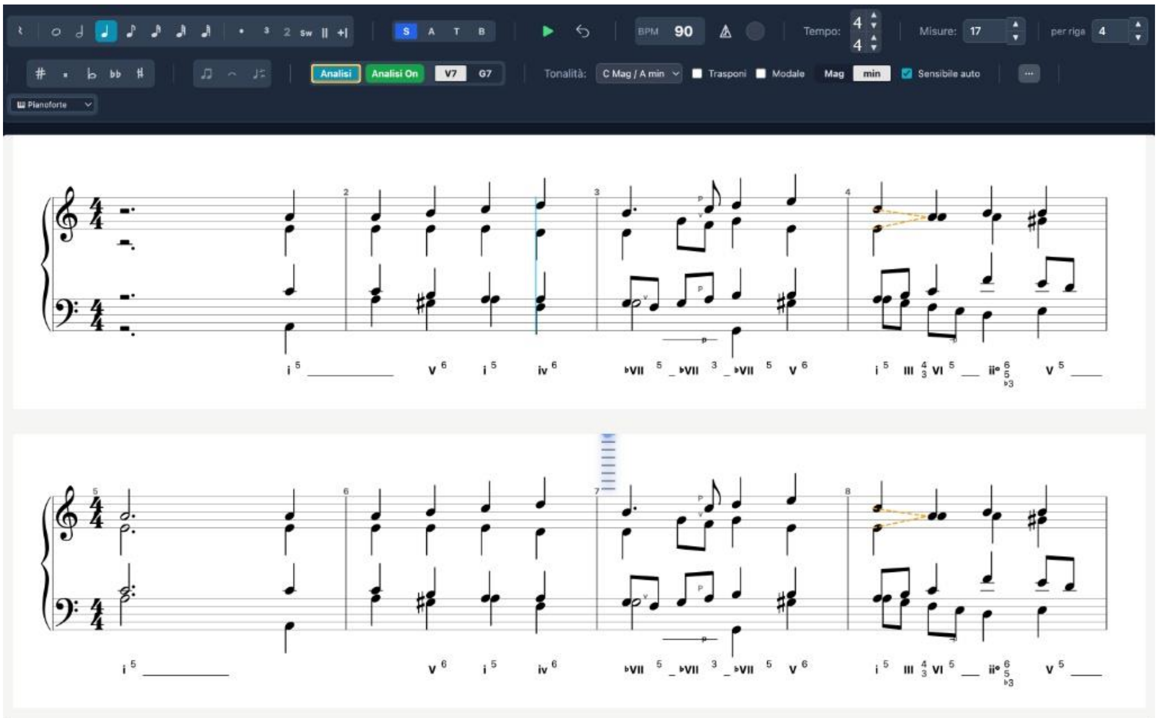


Figure 10 — Score with active harmonic analysis

7.2 The pipeline

The diagram below illustrates the full flow from note entry to label rendering.

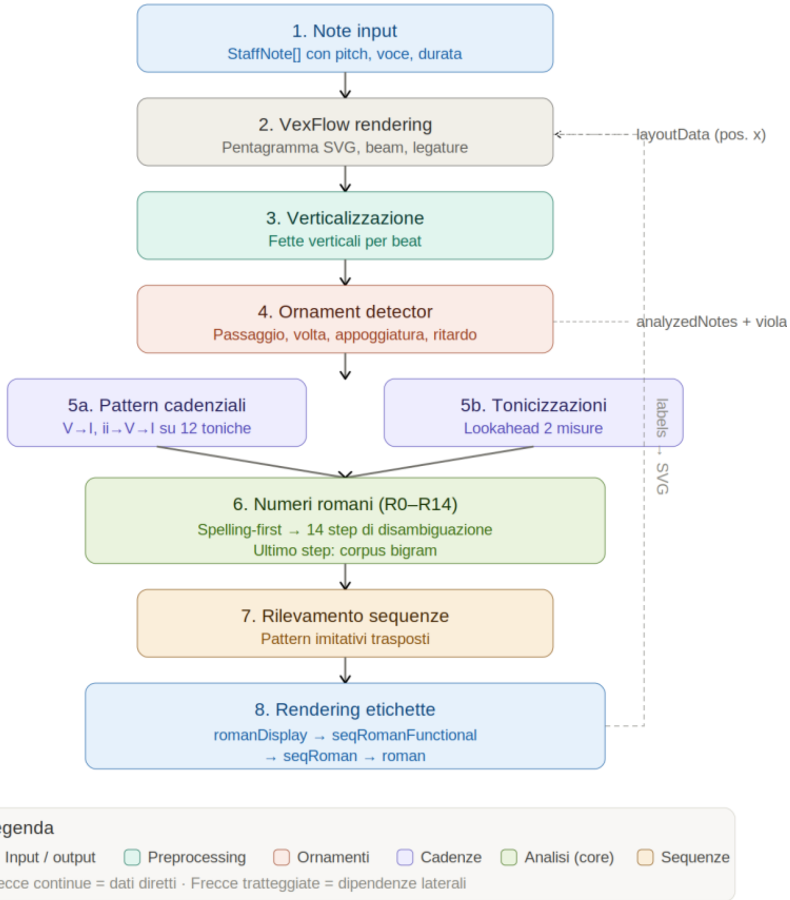


Figure 11 — Harmonic-analysis pipeline: 8 blocks

- **1. Verticalisation:** grouping of simultaneous notes.
- **2. Ornaments:** identification and separation of non-chord tones.
- **3. Chord identification:** spelling-first analysis, candidates ranked by plausibility.
- **4. Roman numeral (R0–R14):** 14 disambiguation steps; the last one uses the statistical corpus.
- **5. Cadential patterns:** recognition of cadences and tonicisations.
- **6. SATB rules:** over 40 voice-leading rules.
- **7. Sequences:** recurring melodic patterns (optional).
- **8. Rendering:** labels below the staff (and symbols above, if enabled).

7.3 Recognised chords

Triads

Type	Example (C)	Intervals
Major	C–E–G	1–3–5
Minor	C–E♭–G	1–♭3–5
Diminished	C–E♭–G♭	1–♭3–♭5
Augmented	C–E–G♯	1–3–♯5

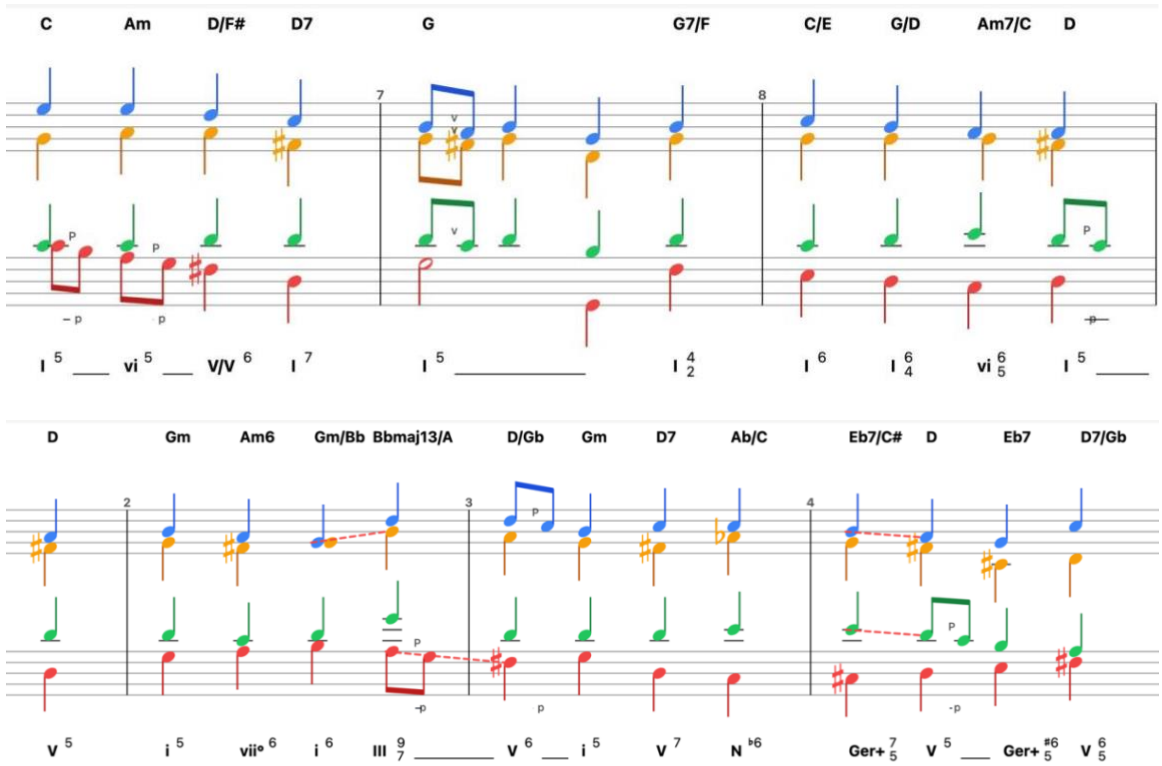
Sevenths

Type	Example (C)	Intervals
Dominant 7	C–E–G–B♭	1–3–5–♭7

Type	Example (C)	Intervals
Major 7	C–E–G–B	1–3–5–7
Minor 7	C–E♭–G–B♭	1–♭3–5–♭7
Diminished 7	C–E♭–G♭–B♭♭	1–♭3–♭5–♭♭7
Half-diminished	C–E♭–G♭–B♭	1–♭3–♭5–♭7

Special chords

- **Augmented sixths:** Italian (It6), French (Fr6), German (Ger6).
- **Neapolitan (♭II):** major triad on the lowered 2nd degree.
- **Dominant ninths:** V9, V7♭9, V7♯9.



Figures 12-13 — Coloured voices with symbols and Roman numerals, ornaments (P, v)

8. Non-chord tones

Harmony Tutor automatically recognises non-chord tones and excludes them from the chord.

Type	Marker	Feature
Passing tone	P	Stepwise between chord tones, weak beat
Neighbor tone	v	Steps away from and back to the real note
Appoggiatura	A	Dissonance on a strong beat, resolves by step
Anticipation	N	Anticipates a note of the next chord
Escape tone	S	Leap + stepwise resolution
Suspension/Retardation	R	Held note, resolves downward
Changing tone	C	Ornament with a leap from the dissonance

Suspensions are classified as 4–3, 7–6, 9–8, 2–1. Manual marking uses the Option (Mac) or Alt (Windows) key + the corresponding letter.

9. Rules and violations

The analysis panel flags violations with coloured badges. Every flag is expandable with a description and a tip.

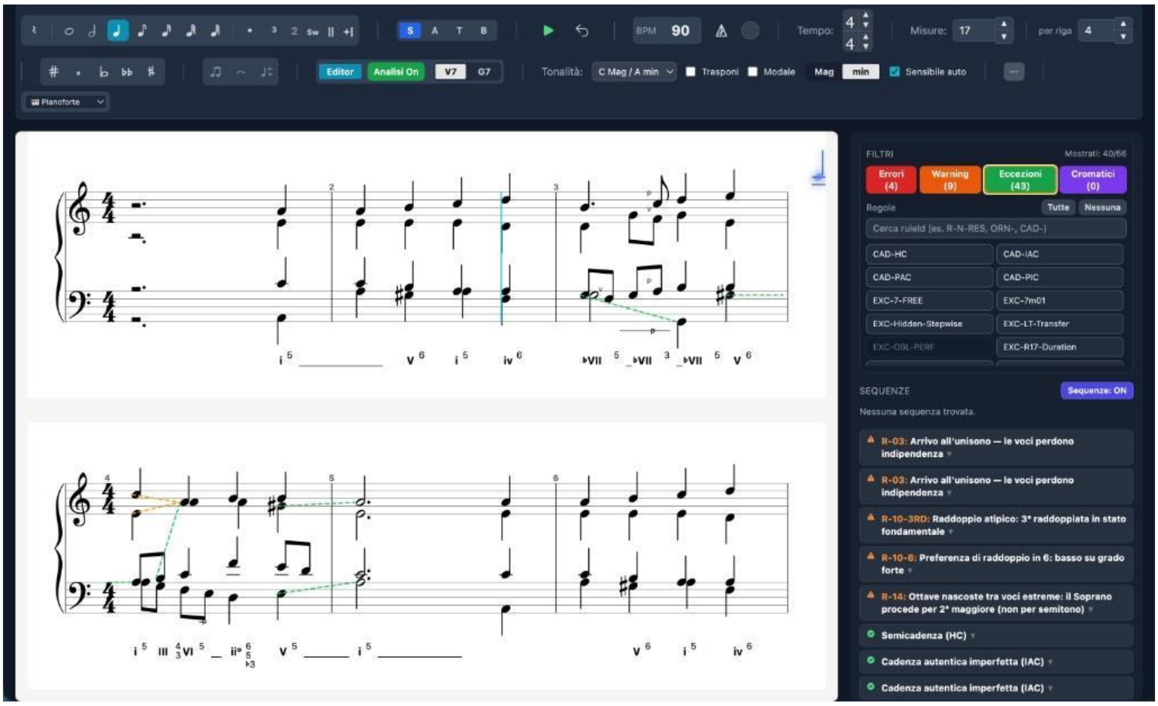


Figure 13 — Analysis panel with filters, violations and cadences

9.1 Severity levels

Colour	Severity	Meaning
Red	Error	Serious violation of fundamental rules
Orange	Warning	Situation to avoid, not always forbidden
Green	Exception	Special case tolerated by tradition
Purple	Chromatic	Advanced chromatic harmony

9.2 Errors (red)

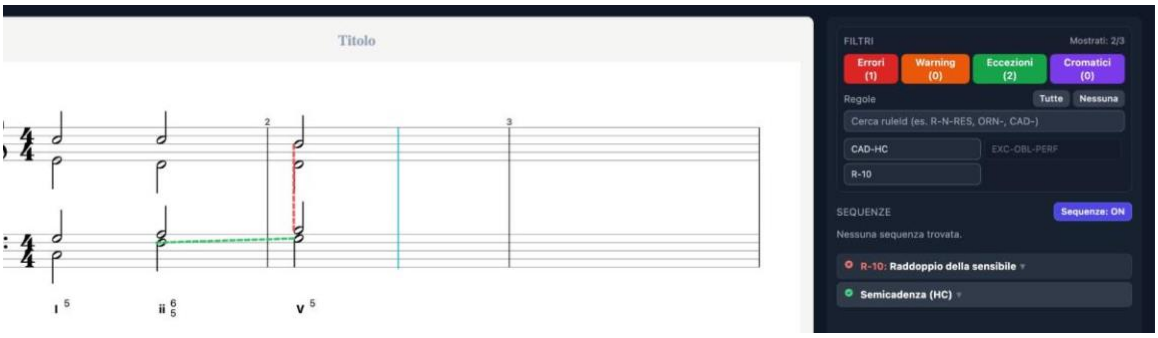


Figure 14 — Example error R-10 (doubled leading tone)

Code	Rule
R-01	Parallel octaves/unisons
R-02	Parallel fifths
R-04	Voice crossing
R-07	Leading tone does not resolve
R-09	Chromatic false relation
R-10	Doubled leading tone

Code	Rule
R-12	Seventh does not resolve

9.3 Warnings (orange)

Code	Rule
R-05	Hidden fifths/octaves
R-08	Excessive spacing S–A / A–T
R-13	Parallel motion in all voices
R-15	Wide leaps in an inner voice
R-16	Harmonic syncopation
R-17a/b/c	Problematic melodic leaps
R-RANGE	Note out of range

9.4 Exceptions (green)

Special cases tolerated by tradition, flagged with a green badge in the analysis panel (e.g. hidden octaves/fifths between outer voices moving by step).

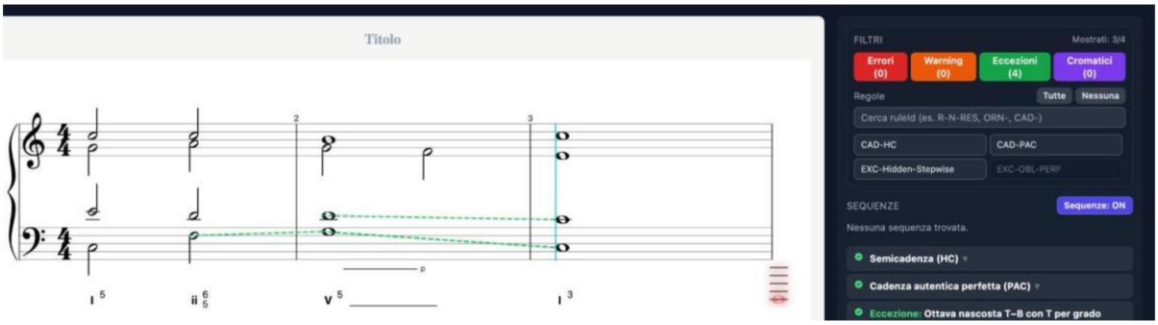


Figure 15 — Exceptions: PAC cadence, HC half cadence, hidden-by-step

9.5 Recognised cadences

Code	Type	Formula
CAD-PAC	Perfect authentic	V → I (root position)
CAD-IAC	Imperfect authentic	V → I (with inversion)
CAD-HC	Half cadence	Phrase ending on V
CAD-PLAG	Plagal	IV → I
CAD-DEC	**Deceptive cadence**	V → vi (maj.) / V → VI (min.) / V → ♭VI (chrom.)

9.6 The analysis panel

- **Filters:** Errors, Warnings, Exceptions, Chromatic checkboxes with counts.
- **Rules:** clickable chips + search field (e.g. “R-01”, “CAD”).
- **Click a Roman numeral:** opens the Explanation card with all the details and alternative readings (≈).
- **Disable rule:** can be silenced individually; persists across sessions.

9.7 Cadential 6/4 → V^6_4

In the cadential progression I^6_4 → V → I, the I^6_4 is automatically relabelled as V^6_4, following the modern functional reading.

- Relabelling happens automatically when the engine detects the V^6_4 → V → I progression.
- Manual overrides (T panel) always take priority.

***Note:** With frequent modulations, check the tonal context before interpreting the V^6_4 label.*

10. Audio and MIDI playback

10.1 Internal playback

Internal playback uses the Web Audio API. Instruments use good-quality sampled timbres (orchestral set, double bass and basses) loaded locally; timbres without local samples fall back to the General MIDI set (FluidR3). The default timbre is Piano. The full palette of selectable timbres (keyboards, strings, basses, woodwinds, brass and tuned percussion) is listed in 11.6 Available instruments.

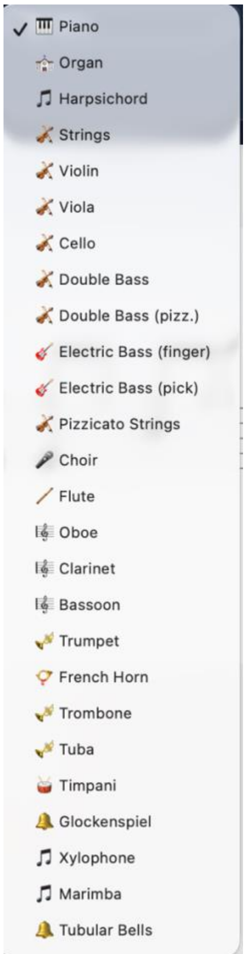


Figure 16 — Available timbres

Dynamics (velocity)

Each note now has a dynamics (velocity) value that affects playback and export. Velocity is captured from MIDI import and live recording, and is used for more expressive playback, for real-time monitoring and for external MIDI output. It is also preserved faithfully in MIDI export.

10.2 Controls

Shortcut	Action
Space	Play / Stop
Enter	Return to start
K	Metronome
+ / –	Tempo
Double-click S/A/T/B	Solo voice

10.3 External MIDI

Harmony Tutor supports MIDI keyboards on input and sending MIDI out to an external instrument or DAW (on Mac, via the IAC Driver). When an external MIDI output is selected:

- the internal piano (Web Audio) is silenced and the sound is monitored by the external instrument/DAW;
- control messages (CC), such as the sustain pedal (CC64), are forwarded, to avoid stuck notes;
- on stop, any notes still active are explicitly turned off (note-off flush).

External playback is velocity-sensitive. Configuration is done from the Preferences panel (MIDI tab) or the side menu.

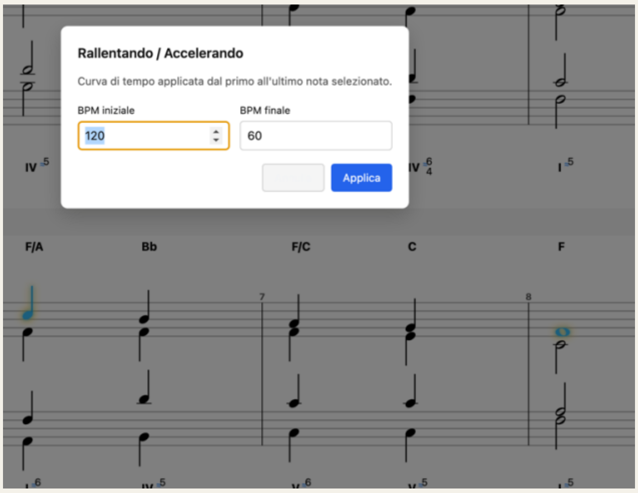
10.4 Tempo curve — Rallentando / Accelerando

The TempoCurve function creates gradual tempo changes over a range of notes. Playback interpolates the BPM linearly between the start and end value. The curve is shown as an overlay on the staff.

How to insert a tempo curve

1. Select the start and end note of the range (Shift+Click for multi-selection).
2. TempoCurve menu, or Shift+Option/Alt+R for a rallentando.
3. Set the start BPM and end BPM.
4. Confirm: the graphic overlay appears on the staff.

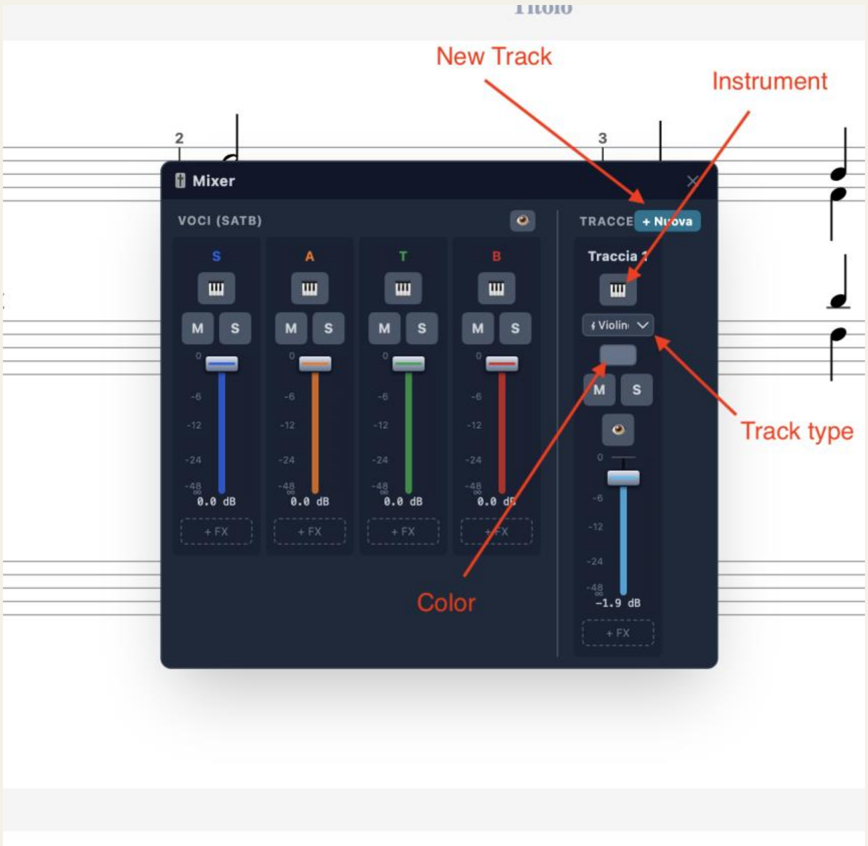
> ***Note:*** *The rallentando/accelerando also applies correctly to the notes of ACC tracks and to loaded files.*



TempoCurve — overlay on the staff with start and end BPM

11. Accompaniment tracks (ACC)

Accompaniment (ACC) tracks are independent of the SATB voices. They are ideal for figured bass, rhythmic patterns or arrangements, without interfering with the harmonic analysis of the chorale.



ACC tracks panel — add track, name, MIDI instrument

11.1 Adding and managing tracks

1. Open the tracks panel from the Mixer or the side menu.
2. Click the “+” button and choose the track type (see 11.4).
3. Assign a name and a MIDI instrument to the track.

ACC notes are visually distinct from SATB notes on the staff, including via the track colour. Right-clicking a Mixer strip deletes the track.

11.2 Mixer

The Mixer is a floating, draggable panel that gathers control of the four SATB voices and all ACC tracks in a single window. It replaces the old sidebar mixer and opens from the Mixer button in the toolbar. The panel is split into two sections: VOICES (SATB) on the left and TRACKS on the right.

Mixer panel — VOICES (SATB) and TRACKS sections

Figure — Mixer: VOICES (SATB) and TRACKS sections

VOICES section (SATB)

One strip per voice: Soprano, Alto, Tenor, Bass. At the top of the section, a visibility button hides or shows the whole SATB block in the editor.

Control	Description
Instrument	MIDI instrument assigned to the voice.
M (Mute)	Mute / unmute the voice.
S (Solo)	Isolate the voice in playback.
Volume fader	Voice gain (0–100%), coloured with the voice colour. Adjustable without stopping playback.
MIDI channel	Outgoing MIDI channel for the external MIDI output: Auto (default) or a fixed 1–16.

TRACKS section (accompaniment)

One strip per ACC track. The “+” button (with its menu) adds a track; right-clicking a strip deletes it.

Control	Description
Track name	Editable label, shown in the track colour.

Control	Description
Staff / clef	Grand staff (treble+bass with a brace) or a single staff (treble, bass, soprano, alto, tenor). Bass and treble are also available as octave-down (8vb) transposing clefs (double/electric bass and guitar): they sound one octave lower than written, while the notation is unchanged.
Colour	Colour picker: vertical band to the left of the staff, tints the track's notes and the fader. SATB voices keep their fixed colours.
M (Mute)	Mute / unmute the track.
S (Solo)	Isolate the track in playback.
Visibility (eye)	Show / hide the track in the editor without removing it.
Volume fader	Track gain (0–100%).
MIDI channel	Outgoing MIDI channel for the external MIDI output: Auto (default) or a fixed 1–16 (Auto: voices 1–4, ACC tracks from 5 up, drums always 10).

> ***Note:*** *Mute, Solo and Volume act in real time during playback (no need to stop and restart).*

> ***Note:*** *The Mixer settings (instruments, volumes, mute, solo, colour and the track staff/clef configuration) are saved in the project file and restored on reopening. Older files without this data start from the defaults (piano, full volume, no mute).*

Adding tracks — the “+” menu

In the TRACKS header, a single “+” button opens a small menu: “New track” (a pitched accompaniment track) or “Drums” (a percussion track). This keeps the fader area uncluttered. The new track appears as a strip on the right.

Master faders (SATB, ACC and global)

Besides the per-channel faders, the Mixer has three master faders: SATB (controls the four voices together), ACC (controls all accompaniment tracks together) and MIX (the global output of the whole mixer). Routing is hierarchical — each channel feeds its group master, and both group masters feed the global master — so the SATB master scales only the voices, the ACC master only the tracks, and the MIX master everything. Each master has its own level meter, and the three values are saved with the project.

MIDI output channel

Every voice and track strip has a MIDI-channel selector for the external MIDI output (toward a DAW such as Logic): Auto (default) or a fixed channel 1–16. With Auto, the four voices use channels 1–4 and the accompaniment tracks are assigned from channel 5 upward; a drum track always goes to channel 10.

Changing the instrument of several voices/tracks at once

The instrument selector in the toolbar normally changes the active voice or track. If you first select notes belonging to several voices (or several ACC tracks), choosing an instrument applies it to all the voices/tracks present in the selection at once — handy for assigning, say, the same string sound to two inner voices in a single step.

Per-channel FX: EQ, Compressor, Pan and Reverb

Every strip — voice or track — has an insert effects chain: Pan (stereo position), a Reverb send, a three-band EQ and a Compressor. EQ and Compressor open in dedicated floating windows, plug-in style with graphs, from the EQ and Comp buttons at the bottom of the strip; a dot on the button shows the effect is active. In the EQ you can drag the three band points directly on the graph (the mouse wheel over the middle point adjusts its width, i.e. the Q); the Compressor shows the transfer curve and a gain-reduction meter. Reverb is a per-channel send: the type (Room, Hall, Plate) and the overall level are chosen on the master. The same EQ and Compressor are also available on the global master (MIX).

Group EQ and Compressor (SATB and ACC)

Besides the inserts on individual channels and on the global master, the two group buses — SATB and ACC — each have their own three-band EQ and Compressor, opened from the EQ/Comp buttons below their master faders. They act on the whole block: the SATB group EQ/Comp shapes all the voices together, the ACC group one all the tracks, before the signal reaches the global master. This is handy to 'glue' and balance the choir and the accompaniment separately. The settings are saved with the project.

Sound bank per voice and track: Orchestral or GM

Next to the instrument selector in the toolbar, a second menu chooses the sound bank of the active voice or track: Orchestral (the local orchestral samples, the default) or GM (the General MIDI soundfont). The two timbres have a different character — the orchestral one has a more realistic attack but can sound a little 'harder', GM is more uniform and softer — and they can be mixed freely, choosing a different one for each voice or track (for example GM on the inner parts and Orchestral on the melody). The choice is saved with the project.

Multiple channel selection and group actions

At the top of every strip there is a selection dot: click it to select the channel, and the strip is highlighted with a coloured border. Shift-click a second channel to also select every channel between the two. When at least one channel is selected, an action bar appears at the top of the mixer with the group actions: Mute and Solo (toggle all selected channels together), Delete (removes the selected ACC tracks — SATB voices cannot be deleted) and Deselect. In addition, dragging the fader of a selected channel moves all selected faders together by the same amount, preserving the group's relative balance.

Reordering and copying track settings

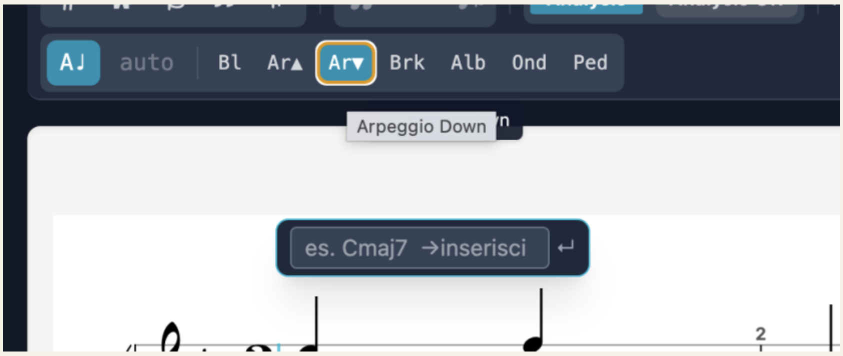
ACC tracks can be reordered from the right-click menu on the strip (Move left / Move right): the order changes both in the mixer and in the score. Also from the right-click menu, Copy settings and Paste settings transfer the clef and the whole effects chain (EQ, Compressor, Pan, Reverb) from one track to another, without touching the notes or the volume.

11.3 Accompaniment patterns

When the active staff is an ACC track, the toolbar shows the buttons for the chord-entry pattern. The chosen pattern determines how the inserted chord is written.

Button	Pattern	Description
Bl	Block	Full simultaneous chord.
Ar▲	Arpeggio up	Notes scaled upward.
Ar▼	Arpeggio down	Notes scaled downward.
Brk	Broken	Boom-chick: alternating bass and chord.
Alb	Alberti	Alberti bass pattern.
Ond	Wave	Up-and-down motion: ascending–descending.

Ped (Pedal / Let Ring) — unlike the previous patterns, this is not a way to insert the chord but an action applied to already-selected notes: it makes them ring until the next attack (pedal effect).



ACC pattern toolbar — Bl, Ar▲, Ar▼, Brk, Alb, Ond, Ped

11.4 Accompaniment track types

When creating a track with “+ New”, you can choose between several types, depending on the writing complexity needed:

Track type	Description
Single staff (Treble, Bass, Soprano, Alto, Tenor)	A single staff in the chosen clef. Suitable for melodic lines, basses, single-voice parts.
Grand staff	Double staff (treble + bass) with a brace, for two-staff writing.
Grand staff with voices	Double staff with the full voice management of the SATB (see below).

Grand staff with voices

This is the most articulated track type and the only one that allows true polyphonic piano writing. It reuses the same engraving engine as the SATB, applied to a single renamable track. It has four independent voices, distributed across the two staves:

Voice	Staff	Stems
Voice 1	Treble (right hand)	up
Voice 2	Treble (right hand)	down
Voice 3	Bass (left hand)	up
Voice 4	Bass (left hand)	down

Because the voices are independent, each can have its own rhythm (polyrhythm) and contain chords: this is the substantial difference from the simple Grand staff, which does not handle rhythmically independent voices.

> ***Note:*** *In practice, a “Grand staff with voices” track is equivalent to a SATB, but as a standalone, renamable accompaniment track: ideal for piano parts or multi-voice writing with rhythmic independence.*

11.5 Copy and paste between tracks

Copy/paste works between any tracks: between different ACC tracks and between SATB and ACC, in both directions. The flow is: copy the selection, choose the destination, paste.

- **Choosing the destination:** a single click on an ACC track's staff (or the SATB staff) sets that track as the active destination and positions the cursor, without inserting anything.
- **Automatic conversion:** pasted notes are adapted to the destination — on an ACC track they take the track's clef (or are split between treble and bass if it is a grand staff); in SATB they take the selected voice and clef.
- **Robust copy:** ACC-track selections and triplets are also copied correctly.

> ***Note:*** *Manual per-note markings (e.g. structural note, ornaments) are preserved during copy/paste between tracks.*

Copying an entire voice (button and shortcut)

To move a whole SATB voice onto an accompaniment track in one operation — instead of copying it line by line — copy the entire voice: right-click the S, A, T or B button in the toolbar and choose “Copy entire voice”, or select the voice and press Shift+Cmd/Ctrl+C. Then paste onto the destination ACC track as usual (Cmd/Ctrl+V, then click the track’s staff). This is the fastest way to send each of the four voices to a different instrument for a wider orchestration.

Usage tip — from chorale to orchestration

One of the most effective uses of copy/paste between tracks is to use the SATB as the harmonic foundation for an arrangement. The typical flow: write and correct the four-part chorale using the real-time analysis, until the harmony is valid and the part-writing is flawless; then copy the individual voices onto the accompaniment tracks, assigning each one an orchestral destination (for example the four voices to a string section). This way the parts enter the orchestral texture already correct in terms of harmony and voice-leading: the checking work done on the chorale is not lost, but becomes the solid base on which to build the orchestration.

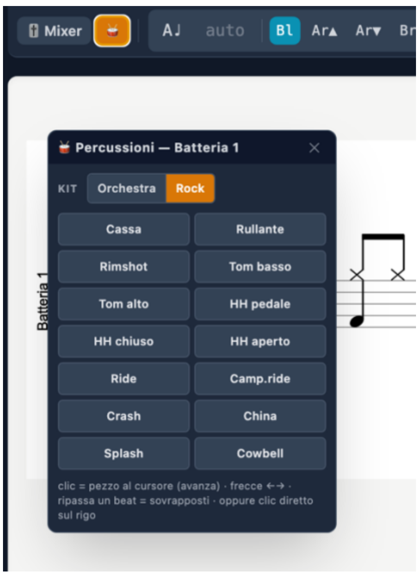
11.6 Available instruments

For every SATB voice and every accompaniment track you can choose the timbre from the instrument button in the Mixer. The sound palette has been expanded with sampled orchestral timbres, grouped by family:

- **Keyboards: Piano, Organ, Harpsichord.**
- **Strings: Strings (section), Violin, Viola, Cello, Double Bass, Pizzicato strings, Choir.**
- **Basses: Double Bass (pizzicato), Electric Bass (finger), Electric Bass (pick).**
- **Woodwinds: Flute, Oboe, Clarinet, Bassoon.**
- **Brass: Trumpet, Horn, Trombone, Tuba.**
- **Tuned percussion: Timpani, Glockenspiel, Xylophone, Marimba, Tubular bells.**

The chosen instrument applies both to internal playback and to external MIDI output. The basses (pizzicato double bass and electric basses) are natural-decay (one-shot) sounds, sampled across the instrument's typical low register; outside that range a note may not sound.

11.7 Drum track (percussion)



In addition to pitched accompaniment tracks, you can add a dedicated percussion track, on MIDI channel 10, to write drum-kit parts using standard notation.

Adding a drum track

- **In the Mixer, TRACKS section, press the button to add a percussion (drum) track.**
- **On creation the floating Percussion module opens automatically, from which you insert the hits.**
- **Alternatively, you can include a drum track right when creating a new project, from the initial-tracks dialog.**

Notation

The drum track uses standard percussion notation: a percussion clef and ╳ noteheads for cymbals. Each kit piece (kick, snare, hi-hat, toms, cymbals…) has its own fixed position on the staff.

The two kits

Two kits are available, selectable per track: Orchestral (default) and Rock. They change the sounds, the piece map and the overall timbre.

Entering hits

- **Percussion module: toggled with the 🥁 button in the toolbar (it appears only when a drum track exists). It shows the kit pieces; click a piece to insert it at the cursor position.**
- **With the mouse: click directly on the percussion staff at the desired point; the hit snaps automatically to the standard line for that piece.**

On playback the track plays the selected kit; on MIDI export it is sent on channel 10. If swing is active, the drum eighth notes also follow the triplet feel.

12. Real-time recording (REC)

Captures MIDI notes from the connected keyboard in real time, with an automatic count-in and post-recording quantisation.

REC button — active state (red) and count-in

12.1 Recording flow

1. Select the destination voice or ACC track.

2. Arm recording with **Shift+R** (requires a visible ACC track).
3. Press **Space** to start: the automatic count-in (default 1 measure) runs and recording begins at beat 1.
4. Play on the connected MIDI keyboard.
5. Press Stop (Space) to finish.
6. Post-recording quantisation is applied automatically.

> ***Note:*** *On ACC tracks the R key toggles note/rest; recording is armed with Shift+R.*

12.2 REC settings

Parameter	Description
Count-in	Number of count-in measures (default 1).
Quantisation grid	Minimum rhythmic value to which notes are snapped.
Overlapping-note trimming	Overlapping notes in the same voice are shortened automatically.

> ***Note:*** *Make sure the MIDI keyboard is configured in Preferences → MIDI.*

13. Import and export

13.1 Import

Format	Extension	Notes
Native	.htp / .json	Complete format
MIDI	.mid	See below
MusicXML	.xml	From MuseScore, Finale, Sibelius

Advanced MIDI import

MIDI import is no longer limited to a simple heuristic voice assignment. When importing a MIDI file (e.g. a piano part) into an accompaniment track:

- **Automatic voice separation:** on a “Grand staff with voices” track, the part is rendered up to two voices per staff (fixed stems: voice 1 up, voice 2 down), without pre-splitting the tracks in a DAW.
- **Automatic clef assignment:** notes below middle C (MIDI 60) go to the bass staff, those from middle C upward to the treble staff.
- **Triplet-aware quantisation,** ties across the barline (a held note survives into the next measure) and correct measure counting, including for ACC-only imports.

13.2 Export

Format	Extension	Notes
Native	.htp	Complete, reopenable
MusicXML	.xml	MuseScore, Finale, Sibelius
MIDI format 0	.mid	All voices on a single track
MIDI format 1	.mid	Each voice on a separate track
PDF	.pdf	For printing
PNG	.png	Screenshot

14. Automatic chorale generator

Generates the four SATB voices from a progression of Roman numerals. Chorale menu → “Insert progression”.

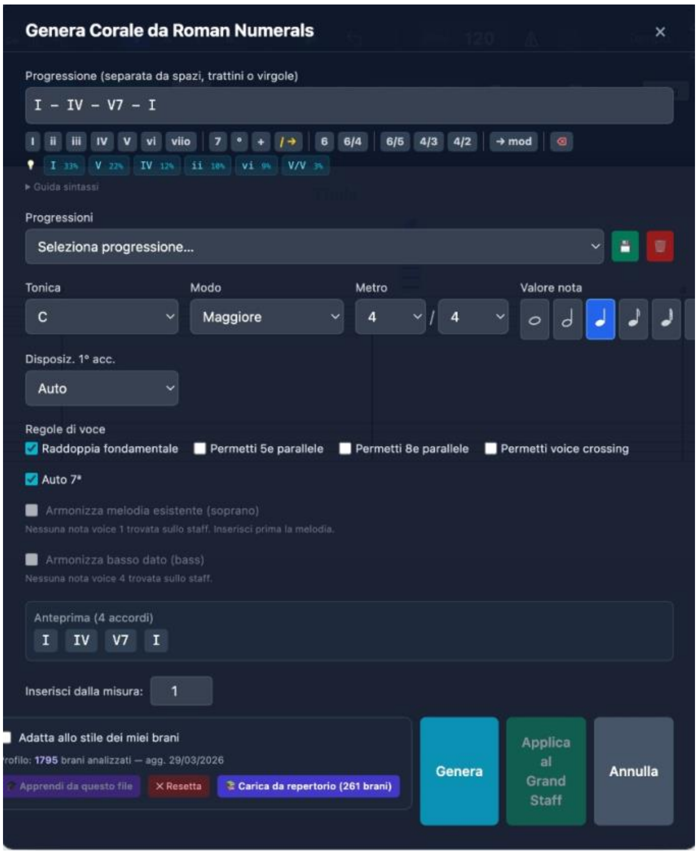


Figure 17 — Chorale generator

14.1 Interface

- **Progression field:** enter chords separated by spaces or dashes.
- **Quick buttons:** I, ii, iii, IV, V, vi, vii°, 7, °, +, 6, 6/4, 6/5, 4/3, 4/2, →mod.
- **Corpus suggestions:** coloured chips with the most likely progressions.

14.2 Settings

- **Tonic, Mode, Meter, Note value:** realisation settings.
- **Voice rules:** double the root, allow parallel 5ths/8ves, voice crossing, auto 7th.
- **Constraints:** given soprano melody or given bass as a fixed constraint.

14.3 Syntax

Syntax	Meaning
I ii IV V vi	Degrees (upper-case=Major, lower-case=minor)
I6 I6/4	Inversions
V7 V6/5 V4/3 V4/2	Seventh and inversions
vii° iiø7	Diminished / half-diminished
V/V V7/IV	Secondary dominants
It6 Fr6 Ger6	Augmented sixths
→G: →f#:	Modulation

Syntax	Meaning
|	Barline (new measure)

14.4 Corpus and learning

The generator uses a statistical system that learns from the pieces entered and analysed. Through bigram and trigram statistics, the software learns the horizontal behaviour of the voices and the most frequent harmonic progressions, guiding the choices of inversion, voice-leading and doubling.

- **Learn from this file:** adds the current exercise to the corpus.
- **Load from repertoire:** imports the pre-computed profile from the corpus.
- **Reset:** restores the default stylistic profile.

15. Teacher mode — Analysis Lock

Lets the teacher prepare exercises by locking the analysis engine and selectively hiding solutions or visual warnings. The lock is saved in the .htp project file: the student can open the file, view the score and play the audio, but the interface hides the elements chosen by the teacher. It cannot be bypassed without the password.

[SCREENSHOT TO BE INSERTED]
Analysis Lock modal — password field and masking options

Figure — Analysis Lock panel

15.1 Activating the lock

1. Help menu → “Analysis Lock”.
2. Set a teacher password.
3. Select the elements to hide.
4. Click “Apply lock”.

15.2 Maskable elements

Option	Effect when locked
Coloured lines and errors	Hides voice-leading flags and disables the violations panel.
Roman-numeral labels	Hides the Roman numerals below the staff.
Chord symbols	Removes the symbols above the staff.
Recognised ornaments	Hides the ornament markers (P, v, A, etc.).
Alternative readings	Removes the ≈ indicator.
Disable export and note copy	Blocks export (MIDI, MusicXML, PDF, PNG) and note copy/paste.

15.3 Unlocking

Click the padlock icon and enter the teacher password. To prevent accidental bypassing, there is no keyboard shortcut for this function.

> ***Note:*** *There is no password-recovery mechanism. Keep it in a safe place.*

16. About the application

Help menu → “About Harmony Tutor…” shows: the installed version, copyright and the website URL.

> ***Note:*** *On Windows this is the only way to check the installed version number.*

17. Preferences and settings

The Preferences panel is organised into six tabs. Settings are saved automatically.

17.1 Editor tab

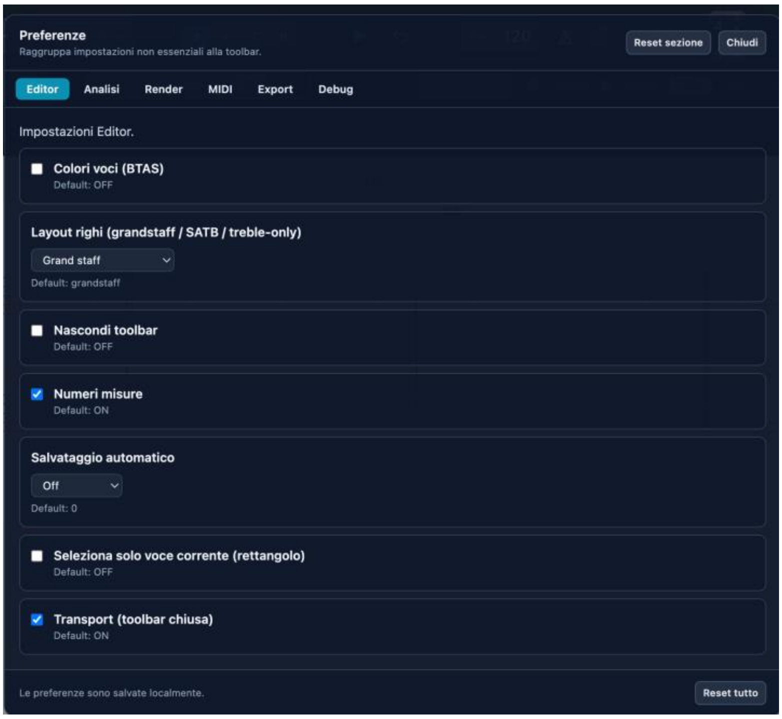


Figure 18 — Preferences: Editor tab

- **Voice colours (BTAS):** distinct colouring per voice.
- **Staff layout:** Grand Staff, Old SATB clefs, Treble-only, Close score, Open score.
- **Hide toolbar / Measure numbers / Autosave:** interface options.
- **Select current voice only:** rectangle selection filtered by voice.
- **Transport:** keeps the transport controls visible with the toolbar hidden.

17.2 Analysis tab

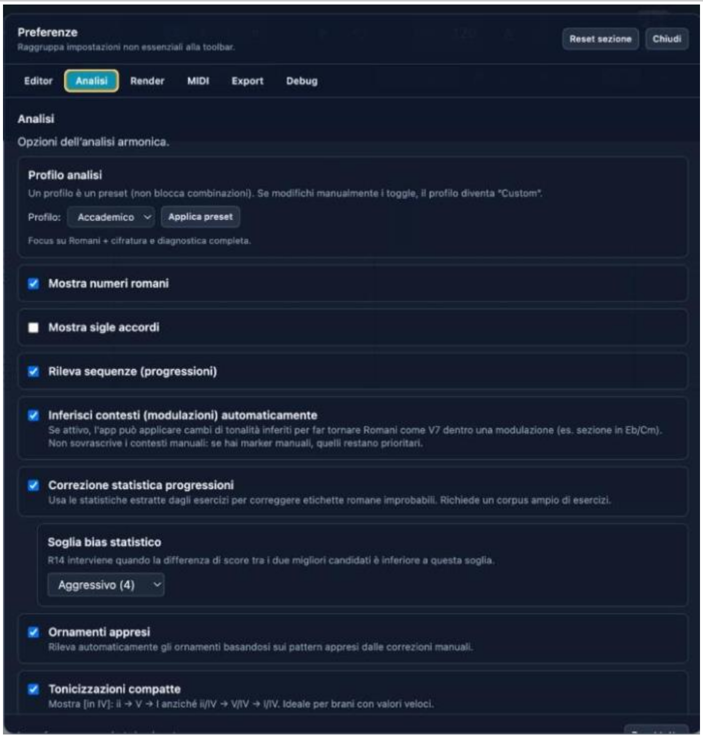


Figure 19 — Preferences: Analysis tab

- **Analysis profile:** Academic preset (default). Customisable.
- **Show Roman numerals / chord symbols:** display toggles.
- **Detect sequences:** recognition of imitated harmonic sequences.
- **Infer contexts:** automatic modulations (manual markers stay priority).
- **Statistical correction:** uses the corpus to correct unlikely labels.
- **Statistical-bias threshold:** Conservative, Moderate, Aggressive.
- **Learned ornaments:** uses patterns from manual corrections.
- **Compact tonicisations / Cadential patterns:** compact format and automatic cadence recognition.

17.3 Render / MIDI / Export / Debug tabs

Render: graphic rendering settings. MIDI: input/output devices. Export: format options (including MIDI format 0 and 1). Debug: advanced diagnostics.

> ***Note:*** *The “Reset all” button restores all settings. “Reset section” only the current tab.*

18. Keyboard shortcuts

18.1 Global

Shortcut	Action
Cmd/Ctrl + N	New project
Cmd/Ctrl + O	Open
Cmd/Ctrl + S	Save
Cmd/Ctrl + Shift + S	Save as
Cmd/Ctrl + I	Import MIDI
Cmd/Ctrl + Shift + E	Export MIDI
Cmd/Ctrl + Z	Undo
Cmd/Ctrl + Shift + Z	Redo
Cmd/Ctrl + P	Print

18.2 Editor

Shortcut	Action
Space	Play / Stop (starts recording if armed, with count-in)
Enter	Return to start
V	Cycle voice
1–7	Duration
# / b / n	Sharp / Flat / Natural
.	Augmentation dot
R	Note / Rest (on ACC tracks)
Shift + R	Arm / disarm / stop recording (with a visible ACC track)
L	Tie
K	Metronome
T	Measure-edit panel
Option/Alt + F	Fermata
Opt + H	Mark the selected note(s) as structural (enters the analysis)
Opt + Shift + H	Collapse the selection (≥2 notes, even an arpeggio) into a single chord
Shift + Opt/Alt + R	Tempo curve (rallentando/accelerando), also on ACC selections and loaded files
Option/Alt + ↑/↓	Move note to treble/bass staff
Option M	Insert close/open score from the playhead onward
Alt + L	Cycle layout
Alt + H	Toggle labels
Cmd/Ctrl + / –	Zoom in/out
Cmd/Ctrl + 0	Reset zoom
Backspace	Delete selection
↑/↓	±1 semitone
Alt + ↑/↓	±1 octave
Shift + Cmd/Ctrl + C	Copy the entire selected SATB voice (then paste onto an ACC track)

***Note:** On different keyboard layouts, the # key may arrive as à, and the . key may produce > with Shift: the app accepts both.*

18.3 Ornament marking (Option/Alt + letter)

Shortcut	Ornament
Option+H	Structural
Option+P	Passing tone
Option+V	Neighbor tone
Option+A	Appoggiatura
Option+N	Anticipation
Option+S	Escape tone
Option+R	Suspension/Retardation
Option+O	Ornamental
Option+C	Changing tone

19. Frequently asked questions

What is the difference between the free version and Harmony Tutor Pro?
The free version includes all features for 10 days. After the trial period, the advanced features (harmonic analysis, chorale generator, export) are disabled. Buying the Pro licence unlocks everything with no time limit.

Can I install the software on a new computer?
Yes. Each licence allows 2 activations. If you change computer, you can deactivate the licence on the old device and reactivate it on the new one.

The analysis shows the wrong chord. What can I do?
Place the playhead at the desired point, press T and enter an override in the panel. Incomplete chords or tonicisations can create ambiguity.

What is CAD-DEC?
The deceptive cadence: V resolves to vi/VI/♭VI instead of I. It is flagged with a green marker in the panel.

What does V^{6}_{4} mean instead of I^{6}_{4}?
In the cadential progression, the I^{6}_{4} is functionally relabelled as V^{6}_{4}. The override is available from the T panel.

How do I insert a rallentando?
Select the range of notes → Shift+Option/Alt+R → set the start and end BPM.

How does the Fermata work?
Select the note → Option/Alt+F. The duration is doubled in playback (default 2×). Repeat to remove it.

How do I add an ACC track?
Tracks panel → “+ New” → choose the type, then assign a name and a MIDI instrument.

Does REC recording work with ACC tracks?
Yes. Select the destination ACC track, arm with Shift+R and start with Space.

Can I import files from MuseScore, Finale or Sibelius?
Yes, via the MusicXML format. Export from those programs and in Harmony Tutor choose File → Import MusicXML. MIDI import is also supported.

What is the difference between MIDI format 0 and format 1?
Format 0: all voices on a single track. Format 1: each voice on a separate track (recommended for editing in a DAW).

Third-party credits and licences

Harmony Tutor — The intelligent tonal-harmony and voice-leading checker
Contact: harmonytutorpro@gmail.com |